

# Vansh Bharat Patil

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## Professional Summary

Computer Science undergraduate (GPA: 9.34/10) with hands-on experience in ML, NLP, and deep learning. Built and deployed end-to-end ML pipelines using Python, FastAPI, and AWS. Research intern at IIT Patna specializing in speech processing and transformer architectures. Multiple national hackathon winner with expertise in knowledge graphs, RAG systems, and MLOps.

## Education

**SVKM's Dwarkadas J. Sanghvi College of Engineering**

2024 – 2028

B.Tech in Computer Science and Data Science — GPA: 9.34/10

## Experience

**AI/ML Developer Intern**

Remote

Xproguard

Jan 2026 – Present

- Developed and optimized **ML models** (Scikit-learn, TensorFlow), improving classification accuracy by **18%**; built scalable **end-to-end data pipelines** reducing data prep time by **40%**.
- Deployed models via **FastAPI** with **REST APIs**, applying **MLOps practices** (CI/CD, model versioning) on **AWS** (S3, EC2).

**Research Intern**

Hybrid

Indian Institute of Technology (IIT) Patna

Jan 2026 – Jun 2026

- Developed **CitationEdge** system for automated claim extraction from scientific papers using **multimodal parsing** (Docling + Vision models) and **hybrid semantic search** (Neo4j + LanceDB); achieved **95%** content extraction and **85%** search recall.
- Implemented **evidence grounding** with verifiability scoring and novelty positioning; **75%** evidence coverage, **72%** automatic verification; deployed as production-ready system with comprehensive documentation.

## Projects

**SignSync — Real-Time Sign Language Communication Platform — MediaPipe, TFLite, FastAPI**

[Live](#) — [Code](#)

- Built a **real-time bi-directional Sign↔Speech system** using **MediaPipe hand landmarks** and optimized **TFLite models**, achieving sub-200 ms inference latency; designed a hybrid pipeline with **keypoint-based static** and **temporal dynamic gesture** classifiers using 21 landmark features and a 16-frame sliding window for robust recognition.
- Integrated **SiGML-based 3D avatar rendering** for natural sign animation with fingerspelling fallback; built **WebSocket/FastAPI** streaming backend and extended the system to **live Jitsi video meetings** with real-time translation overlays.
- Developed a **learning hub** with interactive sign language modules, quizzes, and persistent user progress tracking, making the platform a full end-to-end accessibility and education tool.

**AI Studio for Writers — Knowledge Graphs, NLP, LLM Systems**

[Live](#) — [Demo](#) — [Code](#)

- Built a **knowledge-graph-driven system** (spaCy, Neo4j) extracting structured entities and relationships from 10,000+ narrative documents; integrated **Pinecone vector store** and **RAG pipelines** enabling semantic search and context-aware document retrieval at scale.
- Designed a **context-aware LLM validation engine** leveraging Transformers to detect narrative inconsistencies, character conflicts, and plot contradictions with **92% precision**; deployed **FastAPI** REST backend via **Docker** and **AWS EC2** with CI/CD integration.

**Market Intelligence & Time Series Analytics Platform — Python, Streamlit**

[Live](#) — [Code](#)

- Built a **financial analytics platform** integrating **Finnhub** and **Yahoo Finance** REST APIs for real-time market data; implemented **time-series forecasting** and peer benchmarking models for comparative stock performance analysis across sectors.
- Developed **NLP sentiment analysis pipelines** processing 500+ financial articles daily for market trend prediction; applied **caching strategies** and fault-tolerant design reducing platform load time by **60%**.

## Skills

- Languages:** Python, C, C++, Java, SQL
- ML / AI:** Scikit-learn, TensorFlow, Deep Learning, NLP, Transformers, Attention Mechanisms, LangChain, LangGraph, LangSmith, RAG, Knowledge Graphs
- Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Time Series Analysis, Feature Engineering
- Web / Backend:** FastAPI, Flask, REST APIs, HTML, CSS
- Tools / Platforms:** Docker, AWS (S3, EC2), Git, CI/CD, Streamlit, Gradio, Neo4j, MongoDB, Pinecone, OpenCV, YOLO, Databricks, Dialogflow

## Achievements

- Domain Winner of XAI** of Datahack4.0 organised by DJSCE-S4DS.
- Winner – DevHacks 24hr National Hackathon**, CSI-ACE at Atharva University
- Winner – HackXcelerate 2.0 National Hackathon**, Google Developers Club at VJTI
- Winner – ACE 2.0 National Level 24hr Hackathon**, MPSTME at NMIMS Mumbai
- Second Runner-Up – Spectrum 5.0** (Fintech), SBMP College of Engineering
- Finalist – Rubix'26 National Level Hackathon**, CSI-TSEC at Thadomal College of Engineering
- Finalist – MegaHack 5.0** (Full-Stack Dev.), St. John's College

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